Ying Zhang（章颖）

Ph.D. Student in Computing, Australian National University, Canberra, Australia
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RESEARCH INTERESTS

Data Science, LLM Agents, Computational Social Science
Long-term Vision: Applying computational techniques to generate broad societal benefits.

EDUCATION

Australian National University, Canberra, Australia — May 2026 — Present
Ph.D. Student in Computing
Supervised by Prof. Jing Jiang, Dr. Ahad N. Zehmakan, Prof. Lexing Xie

ShanghaiTech University, Shanghai, China — September 2022 — June 2025
M.S. in Computer Science — Major GPA: 3.75/4.00
Outstanding Graduate (Top 10%)
Thesis *"Intrinsic Structure-Driven Information Acquisition in Data-Scarce Scenarios"*

Communication University of China, Beijing, China — September 2018 — June 2022
B.Eng. in Data Science — Major GPA: 3.85/4.00
Outstanding Graduate (Top 10%)
Selected Coursework: *Principles of Compiler, Operating System, Matrix Analysis, Data Visualization*

PUBLICATIONS (* denotes equal contribution)

[1] **Ying Zhang***, Xiaofeng Li*, Zhaoyang Liu, Haipeng Zhang, "Paths of A Million People: Extracting Life Trajectories from Wikipedia", (ICWSM'25) [Paper & Code]
[2] Hongyuan Tao*, **Ying Zhang***, Zhenhao Tang*, Hongen Peng, Xukun Zhu, Bingchang Liu, Yingguang Yang, Ziyin Zhang, Zhaogui Xu, Haipeng Zhang, Linchao Zhu, Rui Wang, Hang Yu, Jianguo Li, Peng Di, "Code Graph Model (CGM): A Graph-Integrated Large Language Model for Repository-Level Software Engineering Tasks", (NeurIPS'25) [Paper & Code]
[3] Laixin Xie, **Ying Zhang**, Xiyuan Wang, Shiyi Liu, Shenghan Gao, Xing Jiang, Xingxing Xing, Wei Wan, Haipeng Zhang, Quan Li, "Influence Maximization in Temporal Social Networks with a Cold-start Problem: A Supervised Approach", (ICWSM'25) [Paper & Code]
[4] Kai Peng*, **Ying Zhang***, Shuai Ling, Zhaoru Ke, Haipeng Zhang, "Where Did the President Visit Last Week? Detecting Celebrity Trips from News Articles", (ICWSM'24, Spotlight)[Paper & Code]
[5] Xiaofeng Li*, Xiangyi Xiao*, Xiaocong Du, **Ying Zhang**, Haipeng Zhang, "Tracing Footsteps of Similar Cities: Modeling Urban Economic Vitality with Dynamic Inter-City Graph Embeddings", (IEEE BigData'25, Short Paper) [Paper & Code]

UNDER REVIEW

[6] **Ying Zhang**, Hang Yu, Haipeng Zhang, Peng Di, "LLM as Graph Kernel: Rethinking Message Passing on Text-Rich Graphs", [Paper & Code]
[7] Zhongyang Liu, **Ying Zhang**, Xiangyi Xiao, Wenting Liu, Yuanting Zha, Haipeng Zhang, "When Life Paths Cross: Extracting Human Interactions in Time and Space from Wikipedia", [Paper & Code]

RESEARCH EXPERIENCE

ShanghaiTech University, Shanghai, China — Oct 2021 – Jun 2025

Life Trajectory Extraction (Computational Social Science) - Research Assistant — Project Website

- Co-initiated the CeleTrip project (first in the life trajectory series), contributing the core idea and leading execution early; later guided following works.
- Contributed a core member to a trilogy of projects including *CeleTrip* [4], *COSMOS* [1] and *FALCON* [7], extracting spatio-temporal trajectories from millions of Wikipedia biographies.
- Performed empirical analysis of trajectory patterns (e.g. historian mobility) and released both datasets and code for downstream computational social science research.

Ant Group, Hangzhou, China Jun 2024 – May 2026

Graph-based LLMs for Code and Text-rich Graphs (Large Language Model) - Research Intern

- Built pre-training datasets for subgraph reconstruction and implemented the initial GraphRAG modules (Rewriter, Retriever, Reranker), forming the basis for *CGM* [2] on issue-fixing scenarios, which ranked first among open-source models on the SWE-bench Lite leaderboard (May 2025).
- Proposed *RAMP* [6], an LLM-based message passing framework with dual-representation and memory-table propagation for text-rich graphs.

INTERNSHIP

Ant Group, Hangzhou, China June 2024 — May 2026

- **Research Intern — Topics: Code LLM, Text-Rich Graph, Multi-Vector Embedding**
  Working on improving LLM performance in code generation [2] and structured graph reasoning [6]; contributing to data construction, modeling, and evaluation pipelines; exploring multi-vector embedding strategies for semantic retrieval.
- Two patents under review:
  - A Repository-Level Graph-Based Code Large Language Model (No. CN202411612892.0)
  - A Raw Text Anchoring and Neighbor Caching Based Graph Message Passing Method (No. CN202610622272.8)

AWARDS & HONORS

ICWSM Student Travel Award (2024, 2025)
Merit Student ShanghaiTech University (2023, 2024)
Merit Student Communication University of China (2019, 2020, 2021, 2022)
Academic Scholarship Communication University of China (2019, 2020, 2021)

SERVICES

Reviewer for ICWSM (2024, 2025), CIKM (2023, 2024)
Teaching Assistant for CS277 (Introduction to Data Science and Fintech), ShanghaiTech University, 2022 Fall/2023 Fall

SKILLS

- **Programming:** Pytorch Numpy Pandas LaTeX
- **Language:** Chinese (Native), English (IELTS Overall 7.0; L 7.5, R 8.0, W 6.5, S 6.5)